A RELATIVE-CONSISTENCY COUNTERMODEL TO GCH IMPLYING AN $\aleph_2$-SUSLIN TREE

ARGUS OPEN RESEARCH PROJECT

Abstract. We document the relative-consistency theorem of Asperó and Golshani (2024): assuming the consistency of ZFC with a weakly compact cardinal, there is a set-forcing extension satisfying ZFC + GCH together with the special Aronszajn tree property at $\aleph_2$ ($\mathrm{SATP}_{\aleph_2}$), and therefore containing no $\aleph_2$-Suslin tree.

1. Main Relative-Consistency Theorem

Recall that an $\aleph_2$-tree is *special* if it is a union of $\aleph_1$-many antichains, and that every special $\aleph_2$-Aronszajn tree fails to be Suslin because any antichain in an $\aleph_2$-Suslin tree has cardinality at most $\aleph_1$ while a special tree of height $\aleph_2$ cannot be Suslin.

Theorem 1.1 (Asperó–Golshani)**.** *If* ZFC *is consistent with the existence of a weakly compact cardinal κ, then there exists a generic forcing extension $V[G]$ in which $\kappa = \aleph_2$, the Generalized Continuum Hypothesis (*GCH*) holds at all cardinals, and every $\aleph_2$-Aronszajn tree is special. In particular, $V[G] \models$ GCH + "there is no $\aleph_2$-Suslin tree."*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1350 in the domain of set theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *published relative consistency countermodel* deciding the scope: relative consistency from a weakly compact cardinal.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- an unconditional construction inside bare ZFC.

References

[1] Primary problem list and source literature for Result 1350 (set theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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